

Software Requirements Specifications

Blood Donation Management App

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Definition of Terms, Acronyms and Abbreviations

This section should provide the definitions of all terms, acronyms, and abbreviations required to interpret the terms used in the document properly.

Term	Description
ASP	Active Server Pages
RS	Requirements Specifications

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Section

1

1. Introduction

1.1 Purpose of Document

The purpose of this Software Requirements Specifications (SRS) is to define the functional and non-functional requirements for the Blood Donation Management System (BDMS). This system will be used to manage donor registrations, blood requests, donation appointments and hospital coordination. This document is intended for developers, project managers, testers, hospital staff and stakeholders.

1.2 Project Overview

The Blood Donation Management System is a software application designed to efficiently manage the entire process of blood donation from donor registration to blood collection and distribution to hospitals and patients in need. The system aims to reduce manual errors and timely availability of blood.

Software Usage:

The software will be used by blood banks, hospitals and blood donation organizations. Users such as donors, administrators and recipients will interact with the system to:

- *Register and maintain their profiles*
- *Schedule and track blood donation progress*
- *Manage and check blood availability*
- *Generate reports and notifications for blood requirements*

Goals and Benefits:

The system aims to improve the efficiency of blood donation operations, ensures accurate donors and blood records and facilitates rapid communication between donors and recipients.

The system reduced manual work, timely availability of blood, improved management and blood donation services.

1.3 Scope

The system aims to organize the processes of blood donation and storage by

- *Registering donors and maintaining their health history*
- *Managing blood by type and availability*
- *Allowing hospitals and blood banks to request blood*
- *Scheduling and tracking donation appointments*
- *Generating reports for management*
- *Ensuring efficient emergency response*

2. Overall System Description

System Environment

Development Environment:

- *The system will be developed using CPP and OOP.*

Operating Environment:

- **Smartphones:** Android, Ios
- **Internet Environment:** Requires stable internet connection for login, notifications, searching donors and updating data

Anticipated users of the system:**Donors**

- Individuals who want to donate blood
- Age 18+
- Must meet health and eligibility criteria
- Will use the app to receive requests and update donation status

Recipients

- Patients or their family searching for blood
- Hospitals or clinics requesting blood for emergencies
- Will use the app to request blood, search donors and track request status

Admins

- System administration and hospital staff
- Responsible for
 - Verifying donor, recipient accounts
 - Managing emergencies
 - Removing fake accounts
 - Monitoring overall system activity

Constraints

- Requires stable internet connection
- GPS must be enabled for accurate donor search
- Limited battery usage on mobile devices
- Must follow WHO blood Safety guidelines
- Must follow local health rules for donor eligibility
- Must follow data protection laws
- Personal information must be encrypted
- Budget limitations
- Limited admin staff for verification
- Server load capacity

Assumptions

The system assumes that

- Users will provide accurate personal information
- Donors are honest about their health condition
- Users have access to smart phones and stable internet
- Admins will regularly monitor the system
- Hospitals or recipients will update their status after receiving the blood'
- GPS and location services will function properly

Dependencies

The system depends on

- Google Maps for location-based search
- SMS/Email/OTP Service for user verification
- Mobile OS permissions (Location, Internet, Notifications)

2.1 User characteristics

The BDMS will have several types of users with different roles and priorities.

Critical User Classes

I. Donors

Donors are the most important users of the BDMS. They are individuals who donate blood to help those in need. Donors register in the system by providing their personal details such as name, age, blood group, contact information and health status. Once registered, they can update their availability to donate, receive notifications for blood request or emergencies and track their donation history. The active participation of donors is crucial for the system as the whole system depends upon the blood donated by donors and their immediate response to requests.

II. Recipients

Recipients are individuals who request blood. Recipients register in the system by providing their personal details like name, age, blood group to submit blood requests and search for donors. They can track their request status and cancel the request anytime if necessary. They can search for blood by providing blood group they need. They are important for system as core functionality of system depends upon their requests for blood, triggering notifications to donors.

Non-Critical Users

I. Administration

Admins are the system administrators or authorized hospital staff responsible for managing the BDMS. Their responsibilities include verifying users, generating notifications and reports, monitoring emergency blood requests and removing fake accounts. Admins are not actively involved in donor-recipient conversations, but they play a significant role in ensuring the smooth operation, reliability and security of the system. Their involvement is important for maintaining trust.

2.2 Operating environment

Describe the environment in which the software will operate, including the hardware platform, operating system and other software components or applications with which it must coexist.

2.3 System constraints

Identify any constraints or limitations on the system. Constraints may include the following:

- **Software constraints**

Users will need to install the BDMS app because it's not a web-based application.

- **Hardware constraints**

BDMS will be available on any smartphone (android, Ios), laptops and other compatible devices.

- **Cultural constraints**

BDMS is a small level project limited only to Sargodha. No cultural constraints as all languages which are spoken in Pakistan will be available in the system. Users can tailor the language according to their own needs.

- **Legal constraints**

BDMS must follow the ethical rules of WHO and local government.

- **Environmental constraints**

No environmental constraints.

- **User constraints**

System is developed for youngsters, elders and older people. Textual representation is more than graphical representation.

3. External Interface Requirements

This section is intended to specify any requirements that ensure that the new system will connect properly to external components. Place a context diagram showing the external interfaces at a high level of abstraction.

3.1 Hardware Interfaces

Describe the characteristics of each interface between the software and hardware components of the system. This description might include the supported device types, the nature of the data and control interactions between the software and the hardware.

3.2 Software Interfaces

Describe the connections between this system and other external software components (identified by name and version), including databases, operating systems, tools, libraries, and integrated commercial components. Identify and describe the purpose of the data items or messages exchanged among the software components. Describe the services needed and the nature of the inter-component communications. Identify data that will be shared across software components.

3.3 Communications Interfaces

Describe the requirements associated with any communication functions the system will use, including e-mail, web browser, network communications standards or protocols, electronic forms, and so on. Define any pertinent message formatting. Specify communication security or encryption issues, data transfer rates, and synchronization mechanisms.

4. Functional Requirements

This section should contain a textual description of the requirements related to the customer's business. This should contain a list of all the business events related to the business process.

After installation of the app, system will ask the user to create their accounts as donor or recipient.

Name, phone number, email ID, user's photo(optional), location and password will be required.

System will get permission for location access. It'll be mandatory for every user to grant permission so that it will become easy for the system to manage blood donation process.

Location includes country, province, city, hospital etc.

Gender, age, blood group and previous donation history will be provided to the system as per requirements.

Requests will be made and their record will be kept in database for future use.

5. Non-functional Requirements

5.1 Performance Requirements

- **Speed**

System will respond within 2 to 3 seconds, but performance depends upon the speed of internet connection.

- **Precision**

System will be precise; no unreal and unnecessary information will be asked to frustrate the users as this system is built to facilitate the users.

5.2 Safety Requirements

Users should create a strong password. They must keep their personal information private (not necessary to share with other users). When donor and recipient respond to each other's requests and messages, sensitive data isn't important to be shared with each other. Every user should keep the distance from the other. Installed BDMS must be updated after some time for new features.

5.3 Security Requirements

Administration staff will be responsible for the safety of the system and the users. Inactive and fake users will be removed. If any unethical activity is predicted, the culprit will be sued. Fine will be implemented which will be used for the betterment of BDMS. While logging in, system will verify the users by entering CAPTCHA and OTP. Two Factor authentication will be enabled by the users to secure their accounts. Because their negligence will lead them to lose their confidential data.

5.4 User Documentation

Just like all other apps, there will be a help section in main menu in which the whole method of how to use this app will be explained step by step with visual representations. For more details, users can contact customer support in which admins will guide them.

6. Assumptions and Dependencies

List any assumed factors that could affect the stated requirements. These factors are not system constraints, but areas where future changes might drive changes in the requirements. The project could be affected if these assumptions are incorrect, are not shared, or changed.

Also, identify any dependencies the project has on external factors. For example, if you expect to integrate into the system some components that are being developed by another project, you are dependent upon that project to supply the correctly operating components on schedule.

7. References

This section should provide a complete list of all documents referenced at specific point in time. Each document should be identified by title, report number (if applicable), date, and publishing organization. Specify the sources from which the references can be obtained. (This section is like the bibliography in a published book).

Ref. No.	Document Title	Date of Release/ Publication	Document Source
PGBH01-2003-Proposal	Project Proposal	Oct 20, 2003	<Give the path of your Project repository/Folder>

8. Appendices

This section should include supporting detail that would be too distracting to include in the main body of the document.